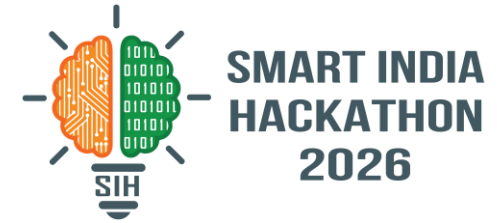


SMART INDIA HACKATHON 2026



TITLE PAGE

PROBLEM STATEMENT ID

SIH26175

PROBLEM STATEMENT TITLE

Single-View Height Estimation and 3D Terrain Visualisation

THEME

Space Technology

PS CATEGORY

Software

TEAM ID

<fill from portal>

TEAM NAME (REGISTERED ON PORTAL)

Dev Up

BUILT AND MEASURED, NOT PROPOSED

3.62_m

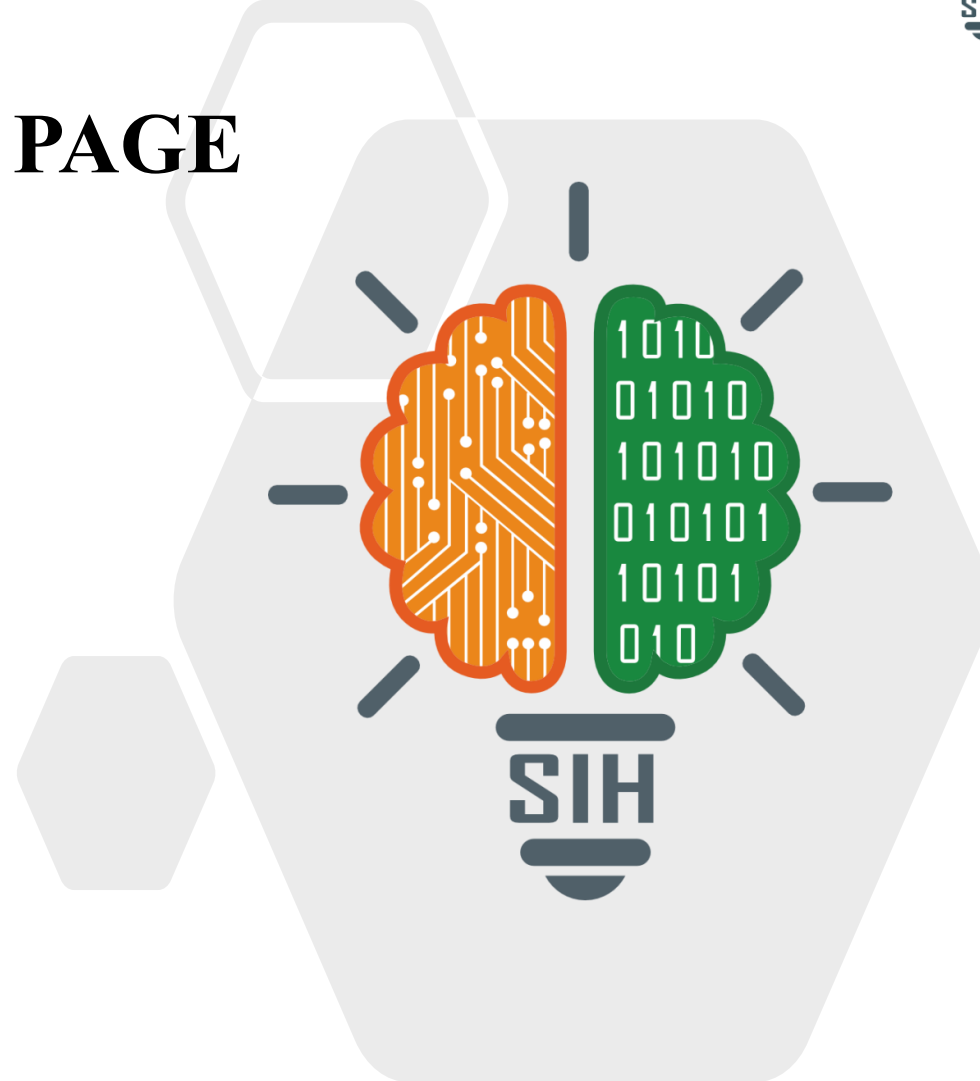
per-building RMSE against
airborne LiDAR, 3,090 buildings

39%

better than the 5.9 m
GlobalBuildingAtlas Asia bar

509_{ms}

per tile on CPU alone,
no GPU required



IDEA TITLE

DepthWizard one satellite image in, a measurable 3D surface out

Height from a single view. No stereo pair, no LiDAR, no second pass of the satellite. Every competing submission promises accuracy. **This one reports it.**

THE PROBLEM

01 Archives with no height

India holds decades of Cartosat single-view imagery. Stereo and LiDAR give height, but cost more and cannot be flown into the past.

02 Answers with no confidence

Single-view height is ill-posed. A model that emits one number implies a certainty it does not have, so an analyst cannot act on it.

03 Checkpoints are not products

The literature ships weights and an eval script. An analyst needs to open a scene, fly through it, and check a height against a reference.

PROPOSED SOLUTION



One image to a metric DSM

Depth Anything V2 fine-tuned on DFC2019 LiDAR. GeoTIFF out, or a relative DSM when the input carries no geospatial metadata.



A confidence map, not just a height

A heteroscedastic head emits per-pixel sigma beside every height. Where it says it is unsure, it is wrong, monotonically.



A navigable 3D scene

three.js flythrough with the image projected onto the surface, drag-to-compare against reference, and a live error map.



Validation built into the product

RMSE, MAE and correlation against LiDAR, reported per terrain class rather than as one flattering average.



Absolute heights, two independent ways

Copernicus GLO-30, or ground control points via a power-law fit that declines rather than degrade a scene.



Deployable, not demo-ware

The model needs no GPU: 509 ms per tile on CPU. The viewer is one 15 MB HTML file, six scenes baked in, opening straight from disk.

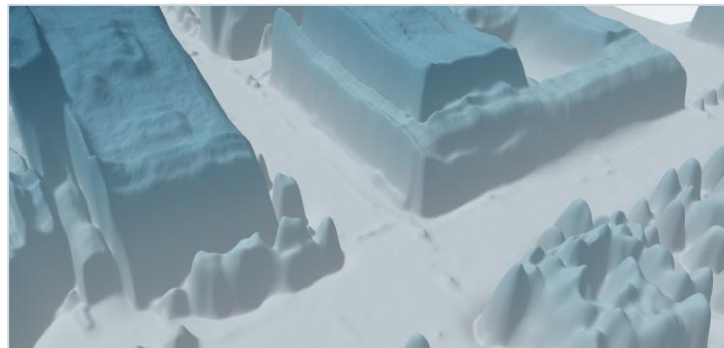
Sikkim, India

967 m of relief across 2 km. Satellite image draped on the recovered surface; terrain from Copernicus GLO-30, everything standing on it is ours.



Omaha, height above ground

The DSM itself. Flat roofs, tree crowns and kerb lines resolved at 0.3 m.



Omaha, model confidence

Per-pixel sigma. The viewer states where the estimate is weak instead of hiding it.



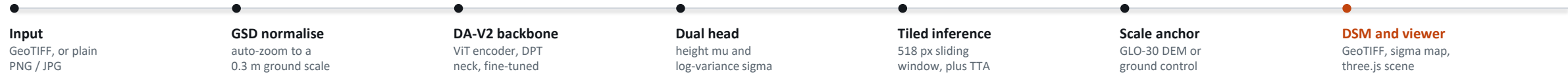


TECHNICAL APPROACH



A monocular depth backbone re-purposed as a metric height regressor, with calibrated uncertainty and a browser renderer.

PIPELINE



TECHNOLOGY STACK

 Python 3.11	 PyTorch	 Transformers	 NumPy / SciPy	 OpenCV
 ONNX Runtime	 three.js r169	 Web UI	 GDAL / rasterio	 Docker

HOW IT IS VALIDATED

Split region-disjoint, not random. Adjacent tiles share buildings, so a random split leaks and flatters the score.

Scored on 80 whole tiles, 11,983,760 valid pixels, 3,090 buildings. Whole tiles, not crops.

Reported per urban / sparse / forested / mixed, plus per-building and per-height-band breakdowns.

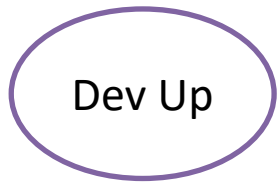
Held back the test split is untouched until the very end, and holds one building above 30 m, so it will flatter us. We will say so.

WHAT SHIPS

An **elevation module** emitting a GeoTIFF DSM plus a per-pixel sigma map, and a **viewer** that opens it, flies through it, and checks a height against reference data. Both run with no GPU and no network.

DECISIONS, AND WHY

- 01 The head was chosen by measurement**
Binned and head-tail-cut heads were both trained and scored. All three land at a 0.43-0.49 slope, so the simplest ships.
- 02 Uncertainty is calibrated, not decorative**
ECE 0.077, sigma-vs-error rank correlation +0.829 on held-out tiles.
- 03 Export is checked, not assumed**
ONNX agrees with PyTorch to 0.0010 cm over 804,972 inputs. int8 costs 0.161 m for a 63% smaller file.
- 04 Resolution is handled explicitly**
auto-zoom reads GSD from metadata and resamples, recovering all but 3.6% of the coarse-input penalty.

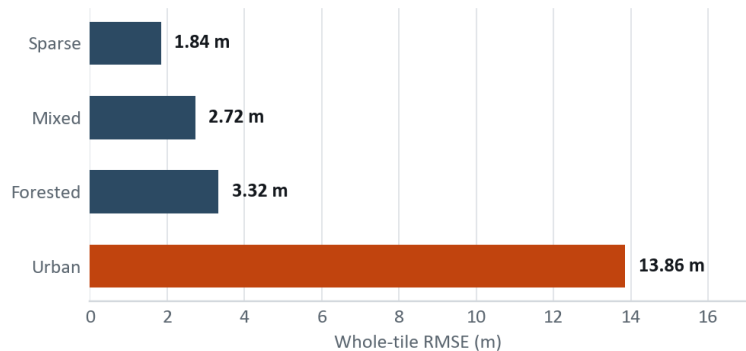


FEASIBILITY AND VIABILITY



The feasibility question is already answered: the system exists. What follows is where it is strong, and where it is not.

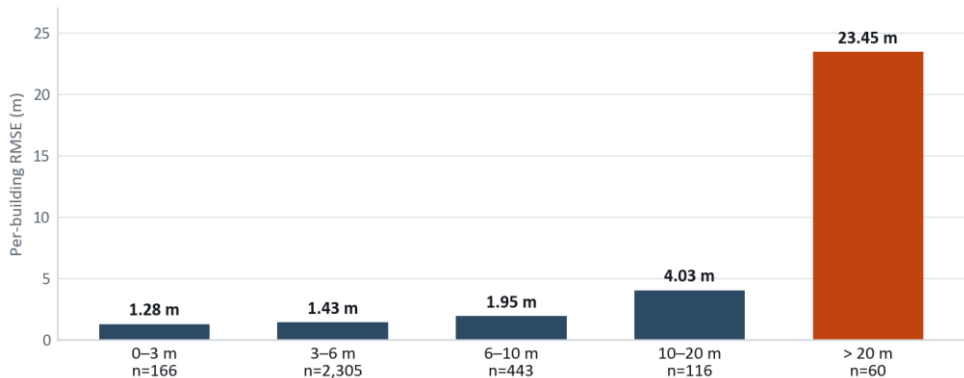
ACCURACY ACROSS THE FOUR LANDSCAPES ISRO NAMES



Three of four classes sit under 3.4 m, and correlation never drops below +0.58. The fourth is the honest problem, and it is one specific thing.

URBAN 13.86 M, DECOMPOSED

79% of all squared error comes from 1.9% of the buildings. Under 10 m, which is 94% of the stock, we are at 1.4 m. Urban RMSE is not a city problem: it is 60 tall buildings dominating a squared metric.



THE TWO REAL RISKS, AND WHAT ANSWERS THEM

01 No Indian ground truth exists to validate against

Our only Indian check is Sikkim against Google Open Buildings: bias -6.94 m on 168 confident footprints. That is model-vs-model agreement, not accuracy, and we report it as such. Cartosat at 0.25-2.5 m is a priced product; free LISS-4 at 5.8 m spans a building in 2-3 pixels. No public Indian scene carries LiDAR truth. This is the honest limit of what we can claim today.

02 Tall buildings compress toward the mean

Proven to be a data problem, not an architecture one: training tops out at 82.8 m while validation reaches 155.3 m, and fine-tuning on the few tall buildings we have moved transfer bias the wrong way (-18.22 to -19.10 m). Answered by a power-law control-point fit that cuts tall-tile RMSE 24.9%, and structurally by adding tall-building LiDAR.

WHAT REDUCES THE INDIAN-DOMAIN RISK BEFORE WE HAVE THE DATA

- Resolution** auto-zoom normalises GSD; measured to recover all but 3.6%.
- Look angle** off-nadir spans 4.8-28.9 degrees in DFC, so degradation vs view angle is measurable on data we already hold.
- Terrain** leakage ruled out. On 31-degree slopes, height-vs-slope correlation is -0.044.
- Licensing** every dependency is free and open, so nothing blocks deployment on ISRO's own imagery once access exists.

Height is the missing dimension in India's satellite archive. Recovering it from imagery that already exists is the whole point.

WHO IT SERVES



ISRO and SAC

Height from single-view Cartosat archives with no new acquisition.



Urban planning

Building heights for FSI checks, sky-view and shadow studies under AMRUT.



Disaster response

Rapid post-event surface models where a stereo pair cannot be waited for.



Telecom rollout

Line-of-sight and tower siting from imagery rather than costly LiDAR survey.



Forestry

Canopy height proxies for biomass estimation and encroachment monitoring.

BENEFITS

Economic

One image replaces a stereo pair: half the acquisition, none of the revisit wait.
No GPU procurement. The quantised model runs on a field laptop.
Existing archives gain a new product without re-flying anything.

Social and civic

Planners and municipal bodies get building heights with no survey budget.
Disaster teams get a surface model in minutes, with its uncertainty shown rather than hidden.
Open, free dependencies keep it adoptable by any state agency.

Environmental and strategic

Canopy and biomass proxies from imagery already being collected.
No new flights means no additional survey emissions.
Reduces dependence on foreign commercial elevation products.

1 image

input needed, where stereo photogrammetry needs two

0 GPUs

509 ms per tile on an ordinary laptop CPU

36.8 MB

the entire model, quantised and running offline

4 terrains

scored separately rather than averaged into one number



RESEARCH AND REFERENCES



Each item below changed what we built. The note beside it says how.

RESEARCH

Depth Anything V2

Yang et al., NeurIPS 2024. The backbone we fine-tune. arxiv.org/abs/2406.09414

HTC-DC Net

Chen et al., IEEE TGRS 2023. Head-tail cut and distribution constraints. arxiv.org/abs/2309.16486

Depth Any Canopy

Ouaknine et al., 2024. The published recipe for adapting Depth Anything to aerial height.

Beta-NLL

Seitzer et al., ICLR 2022. Keeps the mean head learning under a heteroscedastic loss. arxiv.org/abs/2203.09168

IM2HEIGHT, TSE-Net

Prior single-view height estimation. Our baseline for what the field already achieves.

DATA AND REFERENCE SURFACES

IEEE GRSS DFC2019 Track 1

US3D, Jacksonville and Omaha at 0.3 m GSD with airborne LiDAR truth. Training and validation.

Copernicus GLO-30

Free 30 m global DEM, the absolute metric anchor. AWS Open Data.

GlobalBuildingAtlas

Published 5.9 m building-height RMSE over Asia. The external bar this work is measured against.

ISRO Bhoonidhi

bhoonidhi.nrsc.gov.in. Registered; CartoDEM 30 m and LISS-4 are the free tiers.

Google Open Buildings 2.5D

Sentinel-2 derived. Used only as a cross-check over India, never as truth.

TRIED, MEASURED, REJECTED

Ordinal / binned head

Trained and scored. All three heads land at a 0.43-0.49 slope. No gain over regression.

Shadow photogrammetry

Oracle shadows ray-cast from truth reach only r 0.503. The network already reaches r 0.787.

LDS tail reweighting

Premise fails here. Pixels above 20 m are 7.5% of building pixels but already carry 78.8% of error.

Two-model ensemble

Error correlation 0.925. The models make the same mistakes, so averaging buys nothing.

Global de-compression

A linear rescale cannot change correlation by construction. It moves error, it does not remove it.

PROBES WE RAN, AND WHAT EACH ONE CLOSED

01 Tall-building supervision

Can more tall examples move the tail? No. Training holds 166 buildings above 30 m and tops out at 82.8 m; fine-tuning on them moved transfer bias the wrong way, -18.22 to -19.10 m.

03 Guided filtering

Can a post-process sharpen rounded roofs into blocks? No. Guided filtering and morphological toggle contrast each made every metric worse.

04 Where the detail went

What governs detail is the ground area one backbone token covers, not the post-process. That reframed the whole resolution question.

05 Evaluation resolution

Does the model survive Cartosat-class 0.6-1.0 m input? With auto-zoom, all but 3.6% of the coarse-input penalty is recovered.

06 Height compression

Found by eye in the viewer, then measured: ours = 0.473 x truth + 2.66 m, and it cannot be corrected at inference time.